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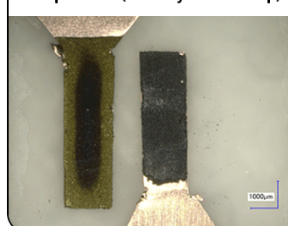
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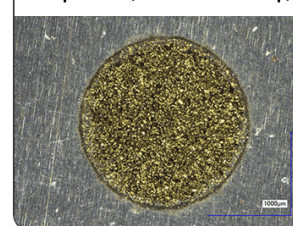
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ELECTROSYNTHESIZED OXIDE SEMICONDUCTORS FOR PHOTOELECTROCHEMICAL APPLICATIONS

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Semiconductors are generally prepared by high temperature methods or by the use of vapor phase and ultra-high vacuum environments. Milder temperatures coupled with the use of condensed media, using water or other solvents (organic compounds or ionic liquids), represent interesting and economic alternatives for the growth of semiconductors in either bulk or thin film form. Electrodeposition is one such preparative methodology in this category, which already has seen widespread acceptance in the metallurgical and microelectronics sectors. On the other hand, electrodeposition of semiconductors has a relatively recent history dating back only to the 1970s. This method is attractive from both fundamental and practical perspectives in terms of applicability to large and irregular device areas and the ability to prepare composite (e.g., metal/semiconductor) structures and unique semiconductor morphologies (e.g., nanotubes, nanodots). This talk will focus on the electrosynthesis and use of oxide semiconductor thin films for photoelectrochemical applications using recent data drawn from our laboratory as illustrative examples.

Compound semiconductors for photoelectrochemical solar energy conversion applications generally involve Group III-V compounds or Group II-VI chalcogenides. On the other hand, oxide semiconductors offer the potential of avoiding precious elements (such as Ga or In) and environmental issues related to disposal or recycling (after the lifetime of the solar device) that beset semiconductors based on As, Se or Te. Nonetheless, the electrodeposition of oxide semiconductor films has an even shorter history and is only now gaining momentum as a serious process candidate. Anodic and cathodic electrosynthesis of metal oxide films will be briefly reviewed in terms of the underlying chemistry/electrochemistry and the mechanistic aspects.

The cathodic electrosynthesis of CdO and ternary Zn-Cd-O thin films from aqueous media will be described. The mechanistic aspects of CdO film formation were characterized by a variety of techniques. The photoelectrochemical behavior of these films will be presented. Oxygenated KCl electrolytes containing ZnCl₂

and CdCl₂ in varying molar ratios were used to cathodically prepare thin films of ZnO, either admixed with CdO or doped with Cd. Remarkably, whether mixed or doped films were obtained, depended crucially on the electrosynthesis potential. Evidence for doping of ZnO with Cd was obtained from X-ray diffraction (XRD), optical transmission, chronopotentiometry, and X-ray photoelectron spectroscopy data. The Cd-doped ZnO samples showed superior photoelectrochemical response relative to undoped ZnO in the ~300-450 nm wavelength range.

Copper (I) oxide has a bandgap energy of 2.0-2.2 eV and a high absorption coefficient over the wavelength span of the solar spectrum. Its component elements are non-toxic and highly abundant. Thin films of Cu₂O were cathodically electrodeposited on Sn-doped indium oxide substrates. The influence of electrodeposition bath pH on grain orientation and crystalline shape was carefully examined using XRD and scanning electron microscopy. An unusual shift in the flat-band potential (spanning ~500 mV) was measured in a non-aqueous electrolyte medium for the various Cu₂O samples obtained from baths of varying pH. A photoelectrochemical cell for H₂ evolution was constructed and studied using the electrosynthesized p-Cu₂O film samples. Data from a variety of electrolytes containing hole acceptors in the anode compartment will be presented along with the photocurrent densities measured for H₂ evolution with visible light.

Composite WO₃-TiO₂ films have been the subject of many recent studies especially because WO₃ and TiO₂ have complementary electronic, optical, photoelectrochemical, and electrochromic properties. Pulsed electrodeposition will be shown to be a useful technique for preparing composite WO₃-TiO₂ films over a wide compositional range. Specifically, a variant of this cathodic electrodeposition strategy that relies on the sequential deposition of WO₃ and TiO₂ from two separate baths will be shown to afford composite films with superior photoresponse relative to counterparts obtained from a single bath.

Finally, our earlier template semiconductor growth studies have been very recently extended to the anodic growth of nanoarchitectures of WO₃ and TiO₂ (as shown by examples in the figure below) starting from tungsten and titanium. The next step in these studies will be to use highly-ordered WO₃ and composite WO₃-TiO₂ nanotube arrays to photocleave water under solar irradiation.

Acknowledgments

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